Pinkie Pie's Balloon Shop

A multi-effect you play as balloons.

Manual for version 1.2.1



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About this manual

The Balloon Shop is a multi-effect you play as balloons. Every balloon is a control, and how full it is shows its value. Your sound goes through a chain of effects in an order you set, the weather changes it, pumps blow up plucked notes or catch sounds into frozen balloons, and a little cloud sets off a sonic rainboom.

This manual covers version 1.2.1. Its middle part, the guide, is the same text you'll find in the plug-in's library: click **help** in the top right corner. Before it come the installation steps, a map of the window and how the sound flows. After it there's a quick reference for gestures, every parameter your DAW can automate, the store's effects, the presets and answers to common problems.

Installing

Download it from the Releases page: `PinkiePiesBalloonShop-macOS.zip` (macOS 11 or later, Apple silicon and Intel), `PinkiePiesBalloonShop-Windows.zip` (Windows 10 or later, 64-bit) or `PinkiePiesBalloonShop-Linux.tar.gz` (x86-64). macOS gets an Audio Unit, a VST3 and a standalone app. Windows gets a VST3 and a standalone app, and Linux adds an LV2.

macOS with Homebrew

```
brew tap femifleming/pinkiepiesballoonshop
brew trust femifleming/pinkiepiesballoonshop
brew install --cask pinkiepiesballoonshop
```

Recent Homebrew asks you to trust a tap before installing from it. If you installed the shop by hand before, add `--force` to the last line so Homebrew can replace those copies. This puts the app in `/Applications` and the Audio Unit and VST3 in `~/Library/Audio/Plug-Ins`. Update with `brew upgrade --cask pinkiepiesballoonshop`.

macOS by hand

1. Unzip `PinkiePiesBalloonShop-macOS.zip`.
2. Copy `Pinkie Pies Balloon Shop.component` to `~/Library/Audio/Plug-Ins/Components` and `Pinkie Pies Balloon Shop.vst3` to `~/Library/Audio/Plug-Ins/VST3`. Put the app anywhere you like.
3. The builds aren't notarised by Apple, so macOS may refuse to open them the first time. In Terminal:

   ```
   xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/Components/"Pinkie Pies Balloon Shop.component" \
     ~/Library/Audio/Plug-Ins/VST3/"Pinkie Pies Balloon Shop.vst3" "/Applications/Pinkie Pies Balloon Shop.app"
   ```

4. Rescan plug-ins in your DAW. It appears under **FreakiePie**.

Windows

Unzip, copy `Pinkie Pies Balloon Shop.vst3` to `C:\Program Files\Common Files\VST3` and rescan. The standalone `Pinkie Pies Balloon Shop.exe` runs from anywhere. SmartScreen may warn you the first time because the build isn't signed: choose *More info*, then *Run anyway*.

Linux

```
tar -xzf PinkiePiesBalloonShop-Linux.tar.gz
mkdir -p ~/.vst3 ~/.lv2
cp -r "Pinkie Pies Balloon Shop/Pinkie Pies Balloon Shop.vst3" ~/.vst3/
cp -r "Pinkie Pies Balloon Shop/Pinkie Pies Balloon Shop.lv2" ~/.lv2/
```

Then rescan in your DAW. The standalone app runs with `"./Pinkie Pies Balloon Shop"` and uses ALSA or JACK.

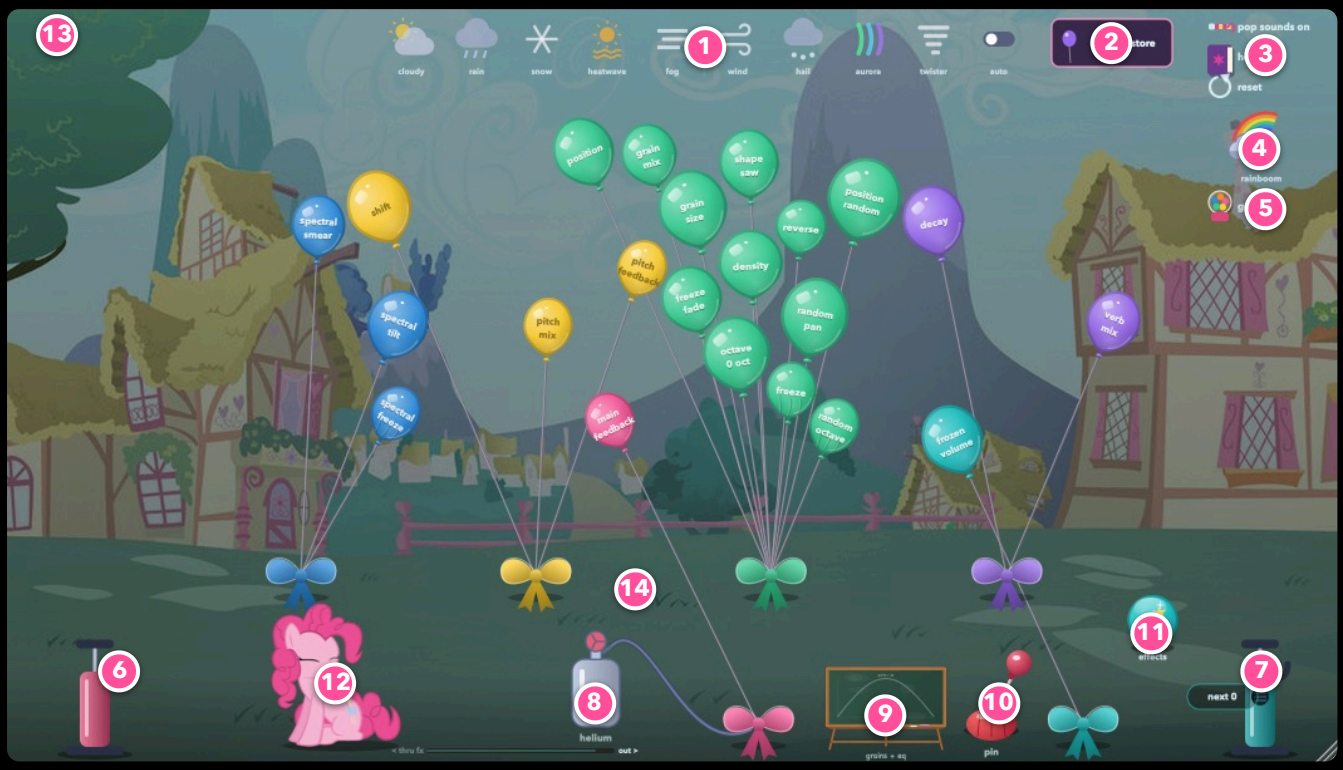
Building it yourself

You need CMake 3.22 or later and a C++20 compiler (Xcode, Visual Studio 2022, or GCC or Clang).

```
git clone --recursive https://github.com/femifleming/PinkiePiesBalloonShop.git
cd PinkiePiesBalloonShop
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release --target BalloonShop_All
```

The window

The numbers on the picture match the list under it. The window can be resized from its corner; everything scales together.



1. The weather row: cloudy, rain, snow, heatwave, fog, wind, hail, aurora, twister and auto weather, all the same size; one that is on is lit up. Right-click the snow or the heat for their fade times.
2. The balloon store.
3. Pop sounds, help (the library) and reset.
4. The sonic rainboom: click for one, right-click for what sets it off.
5. Gumballs: the gumball menu, every control again as a gumball that stays put.
6. The pink pump: blows up plucked notes.
7. The teal pump: catches sounds into frozen balloons. Next to it, the next note's pitch.
8. The helium tank and the canister rail along the bottom.
9. The chalkboard: grain shapes and the master EQ.
10. The pin: drag it to pop a balloon.
11. What the teal pump catches: the effects or the input.
12. Pinkie.
13. Move the cursor here and the card comes down: presets, the scale and the bypass. At the left edge, the order list slides out.
14. The bows on the shelf, one per effect, in the order the sound goes through them. Each holds its bunch of balloons.

How the sound flows

In	Your input, joined by the plucked notes from the pink pump and whatever Trixie is playing.
The effects	The spectral smear, pitch, grains and reverb, the gumball machine and anything bought at the store, in the order on the shelf. The main feedback (the pink loop) feeds the end back to the start.
Bypass	Crossfades the effects out over about 50 ms, leaving your input.
Pops and hiss	Straight out, or into the effects if you choose.
Frozen balloons	The sounds you caught, played as grains. The canister sends them straight out or back round through the effects.
Sonic rainboom	Added on top when one happens, made from what the effects are playing.
Weather	A twister, the wind and the fog change the sound first. Then rain (a lowpass that eases in), helium (a highpass), clouds and thunder. Hailstones, the rainbow and the aurora are added on top, made from what is playing. Last come the heatwave and the snow, which can still freeze all of it.
Chalkboard EQ	The curve you drew.
Leveler and limiter	A gentle leveler, then a look-ahead limiter at -1 dBFS, so it never clips. The look-ahead is 1.5 ms, and the plug-in reports it to your DAW.
Out	Stereo (mono tracks get the two sides mixed).

The guide

The chapters of the library inside the plug-in, with pictures. Each starts on its own page.

Welcome

Every balloon in the shop is a control, and how full it is shows its value.

In short: drag to move, scroll to inflate, shift-drag for fine changes, double-click to reset, click a caught sound to type its pitch, drag the pin to pop them.

Drag a balloon to move it, and let go to watch it float back up on its string. Scroll or swipe over it to blow it up or let some air out. Hold shift and drag up or down to set it by hand (hold alt as well for finer steps). Double-click to put it back how it started.

Balloons with a setting written on them, like shape or octave, step to the next one when you click. Switches like freeze turn on and off with a click.

The bows hold each bunch. Drag a bow to move its whole bunch. The next page shows which colour belongs to which effect.



Dragging a balloon.

Colours

Each bunch has its own colour, and so does its bow.



The library's colour page: which balloon colour belongs to which effect.

The effects

Your sound runs through four effects in a row.

First the spectral smear (the blue balloons), which blurs the sound across time. Spectral freeze holds it still. Spectral tilt leans the smear from dark to bright: up lifts the highs and sinks the lows, down does the opposite, and the middle is flat. The smear stays about as loud either way.

Then the pitch shifter. Shift picks the interval, snapped to the scale chosen on the card at the top left, and pitch feedback sends it round again so it keeps climbing.

Then the grains, and last a reverb that keeps the low end clean. That's the order they start in: the order chapter says how to change it.

Each effect has a mix balloon. When a mix balloon is empty, that effect is out of the way.

Main feedback takes the end of the chain back to the start, so the whole shop can ring on by itself. It is tamed so it sings without blowing up.

Grains

The mint bunch chops your sound into small grains. Grain size sets how long each one is and density sets how many overlap.

Shape is the envelope of each grain. The last shape, chalk, is one you draw yourself on the chalkboard.

Random pan scatters the grains across the stereo field and reverse flips some of them backwards. Octave moves them up or down, or let random octave choose for each grain.

Position picks where in the last four seconds the grains read from, from right now to four seconds back, and position random lets them wander around that spot. Click freeze to hold the sound in place. Freeze fade sets how slowly it slides in and out of the freeze.

These balloons also shape every frozen balloon you catch with the sound pump, even when the effects are bypassed.

The gumball machine

The gumball machine is sold at the balloon store. Buy it and it comes out in an orange gift box with four balloons, and takes its place on the shelf like the other effects; give it back (or pop one of its balloons) and the sound goes straight past it again.

The orange bunch is a stutter that plays with the whole sound in time with your track (or at its own steady tempo when the track is stopped). On each sixteenth note it may grab the moment and do something with it.

Stutter sets how often that happens. Bounce leans it from plain rolls and flams towards bouncing-ball repeats that speed up until they become a pitch, slowing ones that do the opposite, and frozen buzzing grains. Cut-up lets it pull its sound from other steps of the last bar, so the bar gets rearranged like a breakbeat. Sway holds back every second sixteenth, from a gentle swing to a stumble.

These ideas come from Alexander Mesker's study of Autechre's music: the bouncing ball, the event types, restarting patterns and the swing table.

Now and then a pluck from the pop pump bounces too.



The gumball machine, bought and on the shelf.

The order

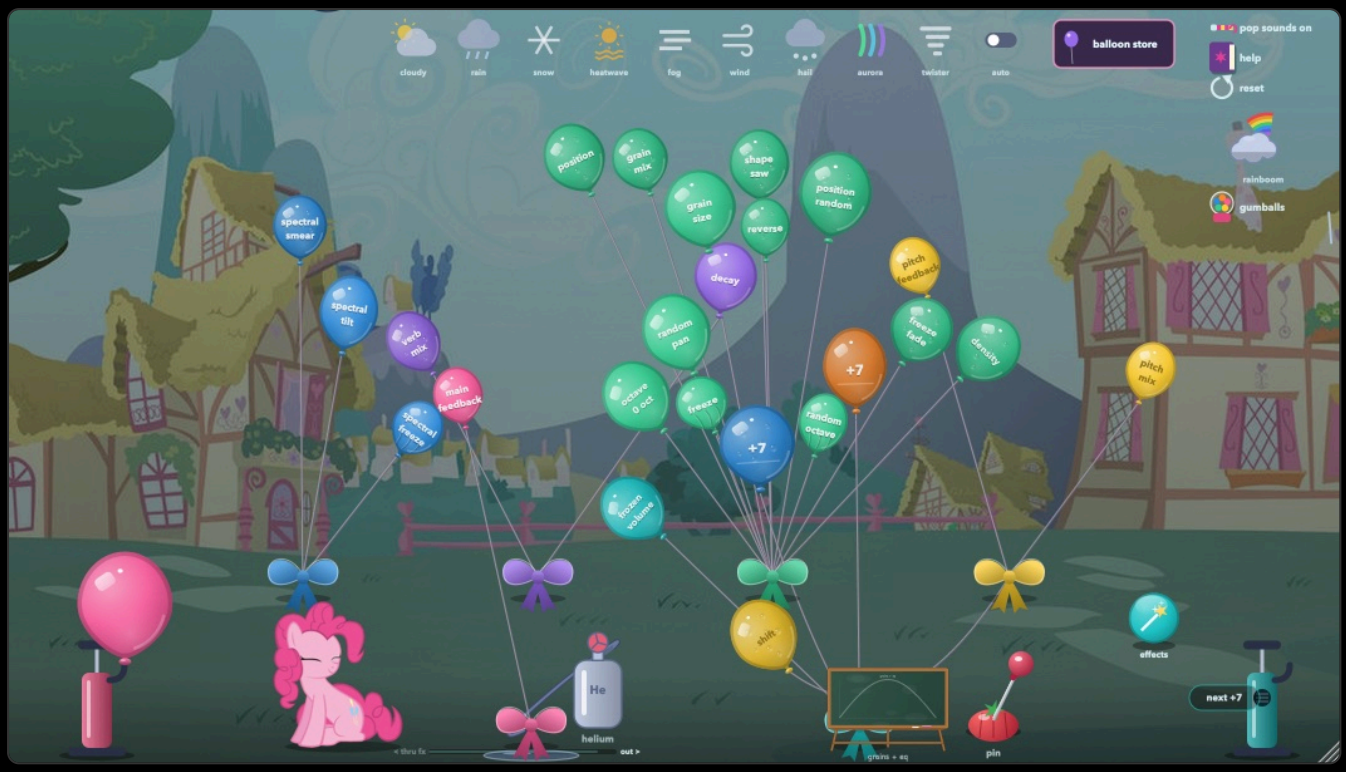
The sound goes through the effects from left to right, from in to out: the spectral smear (blue), pitch (yellow), grains (mint), reverb (lilac) and a gift box for each effect from the store, the gumball machine (orange) among them.

The order list waits just off the left edge of the window. Move the cursor to the left edge and it slides out, with the effects from top (in) to bottom (out). Drag one up or down to move it: the others make room, and when you let go the bows hop to their new places and the sound follows with a tiny dip so it never clicks. The die at the top rolls a new order. The order is saved with your session.

The loop (main feedback) and the caught sounds stay on the floor: the loop wraps round all the effects.



The order list, pulled out from the left edge.



The shelf after a new order.

The pop pump

Hold the pump on the left to blow up a balloon.

While it fills, short plucked saws play on random notes of the scale, slowly and unevenly. Every pluck is folded and panned at random, and its attack is sometimes snappy and sometimes slow. The plucks go through the effects.

Somewhere between 5 and 12 seconds the balloon pops, and every pop is different.

The pop sounds switch at the top right, above help and reset, turns the hiss and the pops on or off. Right-click it to feed them into the effects instead of straight out, so the pops go through the chain with your sound.



Blowing up a plucked note with the pink pump.

Catching sounds

Hold the pump on the right to catch a sound in a balloon. It catches what comes out of the effects, or your plain input if you click the catch token beside it. The balloon starts muffled and opens up as it fills. Let go and it floats away, playing what it caught as frozen grains.

You can keep six at once, and the teal frozen volume balloon sets how loud they all are. The badge by the pump shows the pitch of the next one. Click it, or click any caught balloon, to type an interval like +7 or -5, then press return. Scrolling over a caught balloon retunes it along the scale.

To pop one, double-click it, or drag the pin from the tomato cushion onto it.



Catching a sound with the teal pump.

The canister

Click the helium canister to fill the room with helium. It adds a gentle highpass to the output, and the balloons stretch and float higher.

Slide the canister along its rail to choose where the frozen balloons go. To the right they come straight out. To the left they go back through the effects, so the pitch shifter and the reverb play with them too.



The canister slid towards the effects: the frozen balloons go round again.

Weather

The weather switches are one row along the top of the window, all the same size: cloudy, rain, snow, heatwave, fog, wind, hail, aurora, twister and auto weather. Click one to turn it on and again to turn it off. One that is on is lit up. Rain and clouds together make a storm.

Rain eases a gentle lowpass onto the output and soaks the balloons.

When it's cloudy, clouds drift through the bouquet, and any balloon inside one slowly wanders to new values, with a little tag showing where it has got to. Clouds bring thunder too, which pulls the rain's lowpass down in slow rolls.

In a storm, lightning strikes now and then and drops a random effect from the store into the room for a minute or so. If you like it, tick it in the store and it stays.

Snow freezes everything over the fade-in time: the sound slows until it stops completely and holds, and the balloons ice over and hang still. It stays frozen until you click the snowflake again, then thaws over the fade-out time. Right-click the snowflake to set both times, anywhere from 5 seconds to 5 minutes: drag along a bar or scroll on it, and double-click puts it back to 5 seconds. Leave Pinkie out in the snow for a minute and she catches a cold; she perks up once the snow has gone.

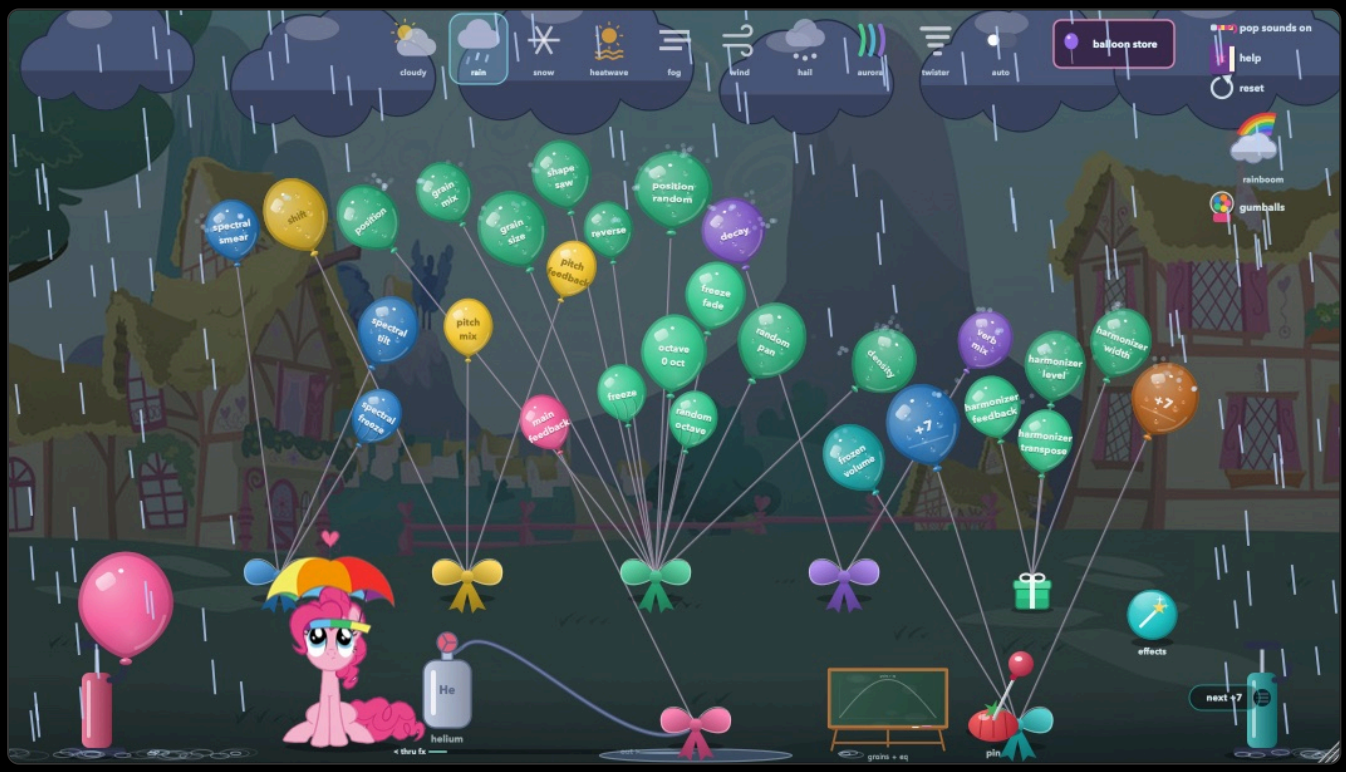
Snow and a heatwave never come together: turning one on sends the other away.

A heatwave melts the sound into a slow smear over its fade-in time (right-click the hot sun to set its fade in and fade out, like the snow's): whatever you play is spread across many seconds, and the held sound keeps drifting towards whatever comes next, like a stretched recording that never quite stops. The top end softens and swells as the hot air wobbles, gently, nothing like the rain. Pinkie can't take it and flops down flat, and if it rains in the heat she's soaked through.

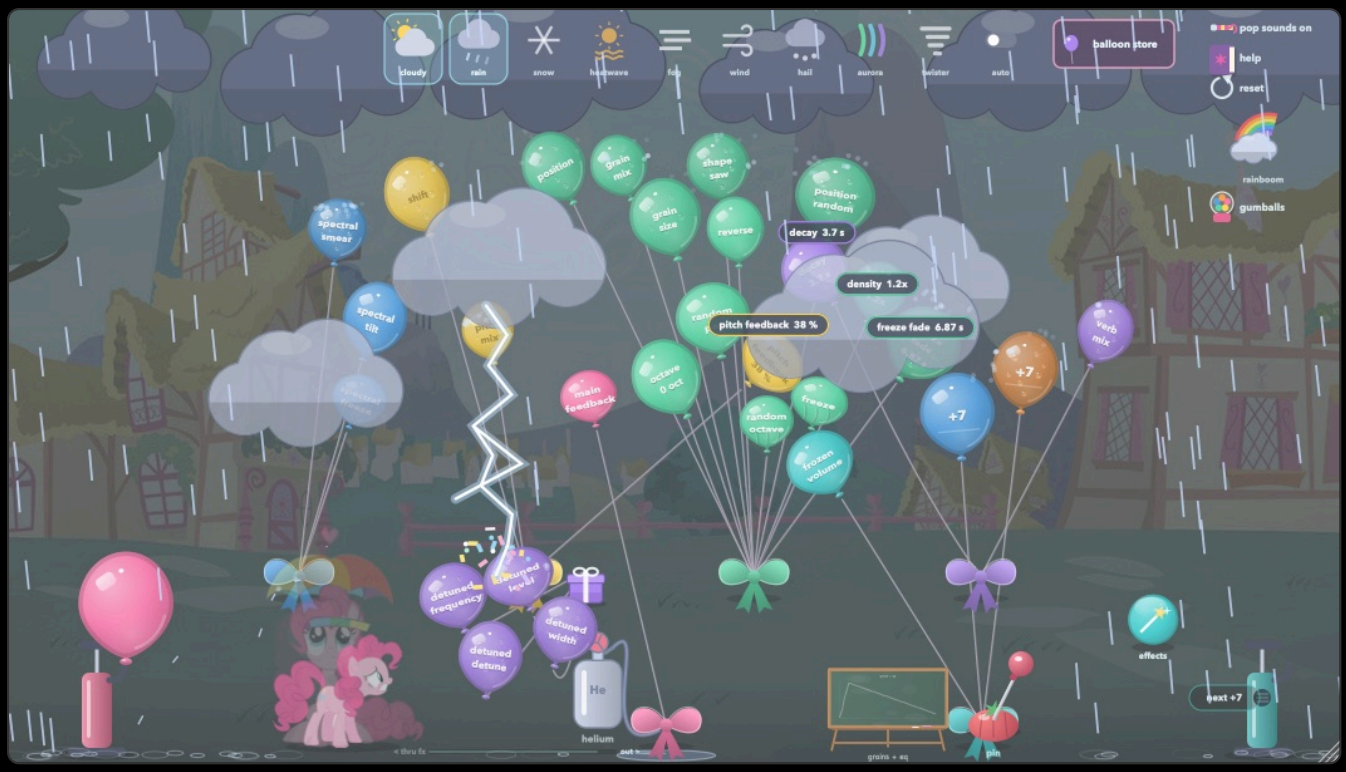
With auto weather on, the sky looks after itself. Every few minutes the weather turns, the way it does outside: clear skies cloud over, the cloud brings rain or a storm or snow, a long hot spell might end in a summer storm. Each spell lasts somewhere between a minute and twelve, and the name under the auto switch shows what the sky is doing. You can still flip the switches yourself; auto carries on from there.



Cloudy: clouds drift through the bouquet and the balloons inside them wander.



A storm: rain and clouds together.



Lightning lends an effect from the store.



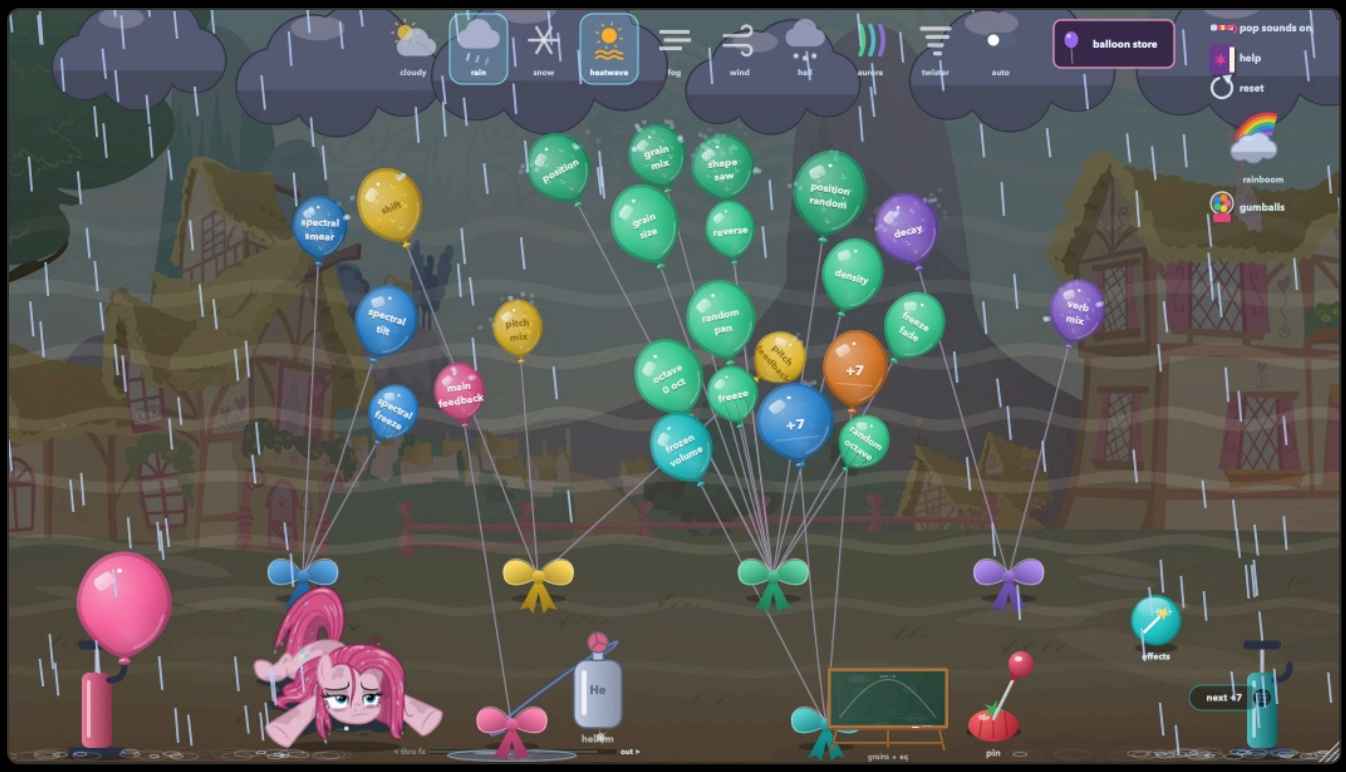
Snow: everything frozen.



A heatwave.



Right-click the snow or the heat for its fade times.



Rain in the heat: Pinkie soaked through.

More weather

Fog, wind, hail, aurora and twister come after the heatwave in the weather row.

Fog blurs every attack. The sound goes through six short echoes too close together to hear apart, so a hit loses its edge without growing a tail, and the highs sink. A pale haze drifts through the shop.

Wind comes in gusts. The sound is read through a delay that keeps drifting, so its pitch wavers a few cents sharp and flat, up to about a fifth of a semitone in a strong gust. Each gust swells the sound a little and leans it to one side, and the balloons lean over with it. The wind blows the fog away: turning one on sends the other off.

Hail throws hailstones. Each one is a tiny slice of the last moment of your sound, hit hard and then bouncing, quicker and quieter, like a ball dropped on a table. Every stone lands somewhere else between left and right.

The aurora plays your sound an octave up into a long reverb. It hangs in four curtains across the stereo field, and each curtain swells and fades by itself.

Twister starts a whirlwind that lasts between 10 and 20 seconds. The pitch climbs a fifth and falls back while the sound whirls from side to side, faster and faster, and the balloons are swept round with it. With wind and clouds both on, one comes by itself every few minutes.

The rainbow has no switch. When the rain stops under a clear sky (no clouds, no snow) after at least 20 seconds of rain, a rainbow stands over the shop for about 25 seconds. It is your sound in seven colours, from red, the lowest, on the left to violet, the highest, on the right, each colour arriving a little after the one before.

Auto weather knows all of these. Fog comes after cloud, rain or snow. Wind can blow up a storm. Hail falls out of a storm, and an aurora follows a clear spell.



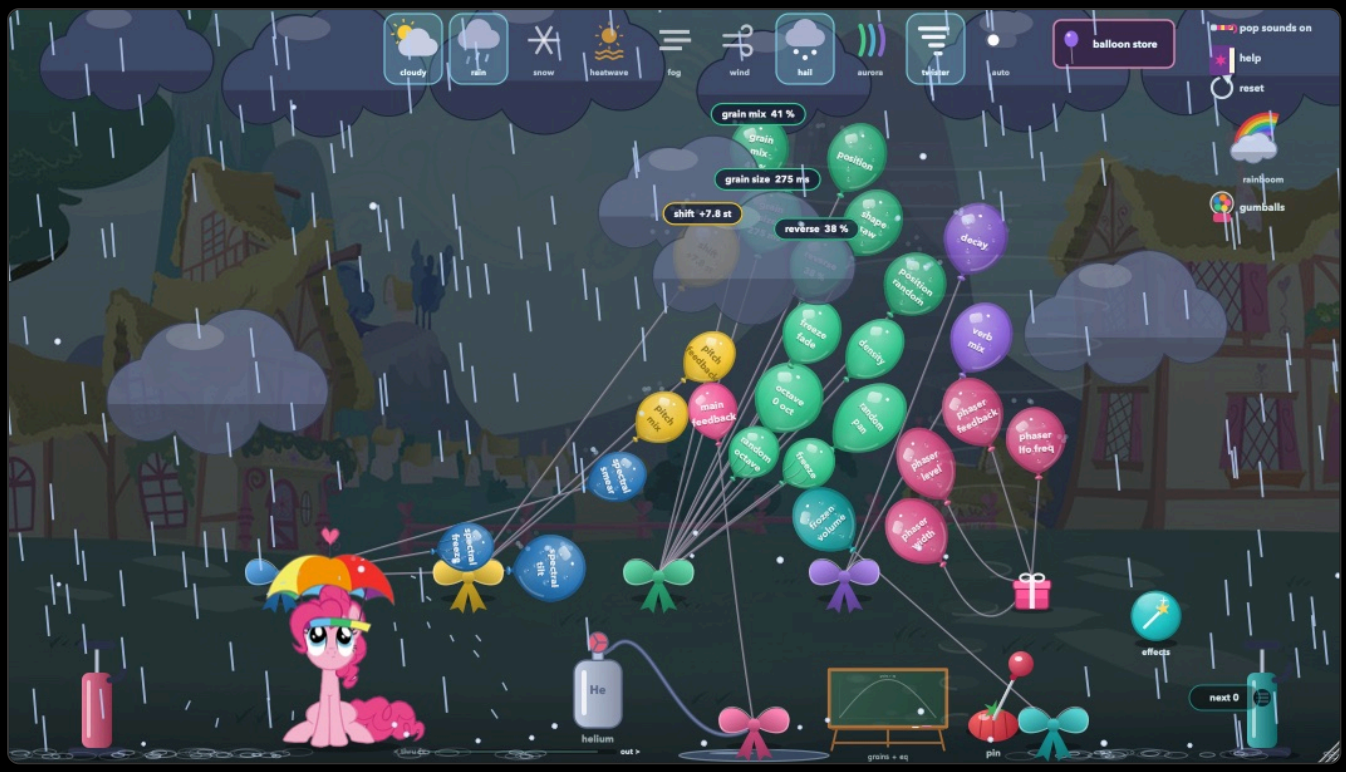
Fog.



Wind: streaks of air, and the balloons leaning over.



A twister.



Hail in a storm.



A rainbow after the rain, and the aurora across the top of the sky.

Sonic rainboom

The little cloud with a rainbow, under help and reset, is the sonic rainboom. Click it and Rainbow Dash dives in from the sky, faster and faster, the rainbow streaming out behind her, until she breaks the sound barrier: a boom and a rainbow at the same moment.

On the way in your sound is squeezed up in pitch, the way a sound rushing towards you would be, with a rush of air under it. The boom has the real shape of a sonic boom, a double crack. Then the rainbow: your sound split into seven colours, red the lowest and violet the highest. They arrive one after another, so they sweep up from the lows to the highs, and each one keeps ringing for the rainbow's length while the rainbow fades from the sky.

The shockwave knocks the balloons back, and the ones nearest it are nudged to new values. Pinkie jumps, amazed.

Fling a balloon hard enough and it breaks the barrier itself: a white cone grows behind it as you get close. Right-click the cloud for what else sets one off, such as a hit in your sound or the rain stopping after a storm. The same menu sets how loud it is and how long the rainbow lasts.

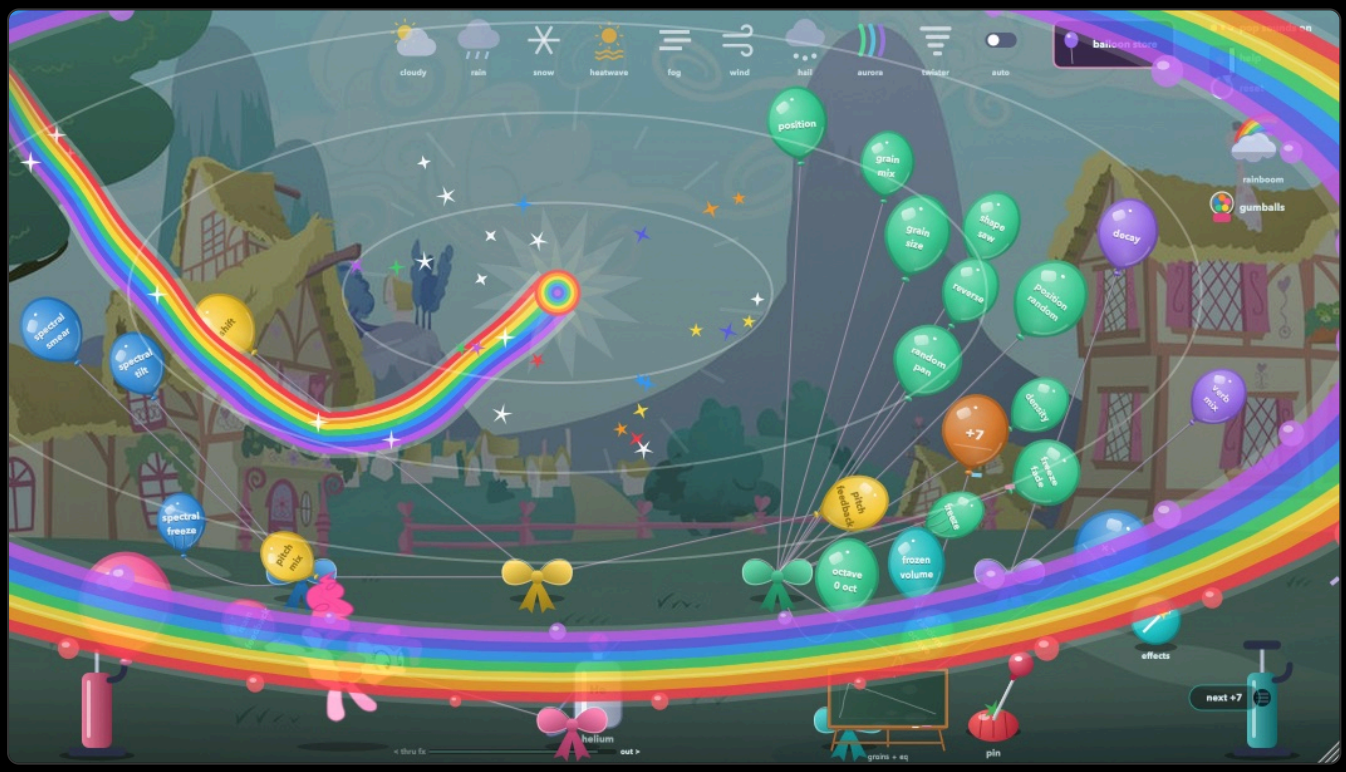
In stereo the colours fan out as the ring opens, red to the left and violet to the right.



The dive: Rainbow Dash speeding in, the rainbow streaming out behind her.



The boom: the starburst and the ring of seven colours. Pinkie jumps, amazed.



The ring opening out; the shockwave has knocked the balloons back.



The rainbow left in the sky, fading over the rainbow's length.

The balloon store

Click the balloon store sign at the top to open the store. Pick an effect and a little gift box drops out of the shop with four balloons tied to it: its two controls, then its level and its width. The box takes a place among the other effects, and the order list moves it about like the rest. Width spreads the effect from mono in the middle to wide stereo.

Pick a bought effect again to give it back, or pop one of its balloons with the pin. Pop everything sends them all back at once.

The tuned effects in the store follow the scale as well. The ring mod multiplies the sound by a tone, and with follow turned up the tone rises as the sound gets louder.



The balloon store.


cloudy
rain
snow
heatwave
fog
pop sounds on
help
reset
rainboom
gumballs
Balloon Store
Reverb
Delay
Disto
Waveguide
Complex Resonator
Degrade
Harmonizer
Clipper
Phaser
Detuned Resonator
Ring Mod
Gumball Machine
pop everything
Disto: yours! click to pop it
position
grain mix
shape saw
grain size
reverse
shift
spectral smear
spectral tilk
pitch mix
pitch feedback
spectral freeze
main feedback
effects
next +7
He
helium
pin

Its shelves.

The gumball menu

Click gumballs, under the rainboom's cloud at the top right, to open the gumball menu. It shows every control in the shop again as a gumball that stays where it is, for when you'd rather not chase a balloon. There is one row for each effect, in its balloons' colour.

Each gumball is a glass ball that fills with colour as its value goes up, with the value written inside. Drag one up or down to turn it (hold shift for fine changes), or scroll over it. Click a switch to flip it. Click a setting like shape to step to the next one, or right-click for the one before. Double-click puts a gumball back how it started. Point at one to read its full name and value.

The gumballs and the balloons are the same controls. Turn one and the other follows, and the balloons still work just as they did.

When an effect's balloons come into the room, bought at the store or dropped by the lightning, its row of gumballs pops in here. When they pop, the row pops out. With many effects bought, short rows share a line and the page scrolls.

Click the cross or press escape to go back.



The gumball menu: a row for each effect.



With the gumball machine and a store effect bought: their rows have popped in.

The chalkboard

Click the little easel on the floor to bring the chalkboard forward. It has two pages, written along the top.

On grain shape, draw across the board to make the life of one grain, from start on the left to end on the right, with loud at the top. Drawing switches the grains to your shape.

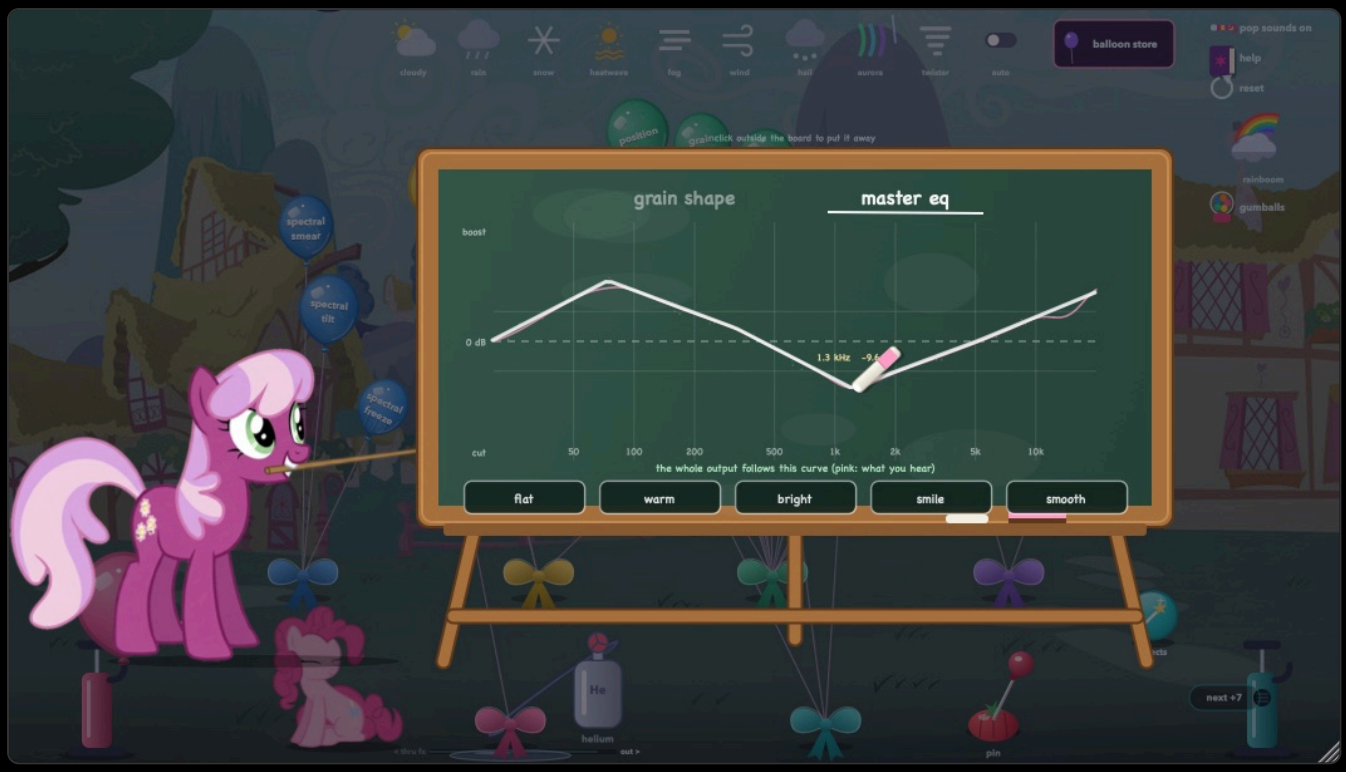
On master eq, draw the tone of the whole output. Above the dashed line boosts and below it cuts, with the lows on the left. The pink line shows what you actually hear. The buttons underneath give you ready-made curves, smooth what you drew, or set it flat again.

On master eq the board goes from a deep cut at the bottom to a big boost at the top, 24 dB each way.

Click anywhere outside the board to put it away.



Drawing a grain shape on the chalkboard.



Drawing the master EQ.

The card

The card holds the presets and the scale. It waits just above the top left of the window: move the cursor up there and it slides down. Every preset changes the picture behind the shop as well as the sound.

The scale decides which notes the pitch shifter lands on. Everything tuned in the shop follows it.

The reset button under the help book in the top right corner brings everything back to normal: the first preset, a flat EQ, the plain grain shape and the effects in their first order. Caught sounds stay where they are.

The switch underneath the card bypasses the effects, the store's ones included. Only your input and the frozen balloons are left, and the grain balloons still shape them.



The card, down from the top left corner.


cloudy
rain
snow
heatwave
fog
wind
hail
aurora
twister
auto
balloon store
pop sounds on
help
reset
rainboom
gumballs
position
grain mix
shape saw
grain size
position random
reverse
decay
density
spectral smear
shift
spectral tilt
random pan
octave 0 oct
+7
freeze
freeze fade
main feedback
spectral freeze
pitch mix
pitch feedback
random octave
frozen volume
+7
verb mix
start
feedback
stretch
useful
scroll or drag along ↔
Sugarcube Corner
Canterlot Chimes
Moonlight Howl
Smokey Mountain Squall
Las Pegasus Neon Loop
Cloudsdale Stretch
Still Lake
Slow Rainbow Falls
Golden Hour

The presets.

Pinkie

Pinkie sits by the pop pump watching the balloons.

Pops make her jump. When it rains she gets her umbrella out (unless it's a heatwave too: then she lies in the puddle, soaked), and helium turns her to crystal.

Leave the shop alone for a while and she falls asleep.

Every so often, about once an hour, the Great and Powerful Trixie barges in. She zaps balloons from her amulet, each one jumping to a new value, then sets everything she can reach at random with one big flash and leaves. Some visits she also plays a strange synthesised noise into the shop's input, which goes through all the effects, with a chirp for every zap. Sometimes she zaps a few effects from the store into the room too, and those stay. She leaves the bypass and the volume alone, and never touches the snow, the heat or auto weather. If you don't like what she did, the reset button under the help book puts everything back to normal.



Trixie drops by.



Leave the shop alone for a while and Pinkie falls asleep.

Quick reference: what your hands do

Drag a balloon	Move it. Let go and it floats back up on its string.
Scroll or swipe on a balloon	Blow it up or let air out (its value).
Shift-drag up or down	Set it by hand. Add alt for finer steps.
Double-click a balloon	Back to its default.
Click a switch or a choice	Switches flip; choices like shape step to the next.
Drag a bow	Move its whole bunch. Along the shelf, it changes the order.
Drag the pin onto a balloon	Pop it (caught sounds, store effects).
Click a caught sound	Type its pitch.
Right-click the snow or the heatwave	Their fade times.
Right-click pop sounds	Pops on or off, or into the effects.
Click the rainboom cloud	A rainboom now. Right-click: what sets them off, level and the rainbow's length.
Fling a balloon hard	Breaks the sound barrier: a rainboom where it's got to (when on throws is ticked).
Click gumballs (top right)	The gumball menu: every control as a gumball that stays put. Drag or scroll one to turn it, click a switch, double-click to reset. Escape closes it.
Click twister	A whirlwind now, for 10 to 20 seconds.
Cursor to the top left corner	The card: presets, scale and bypass.
Cursor to the left edge	The order list. Drag an effect up or down, or roll the die.

Parameter reference

Every control your DAW can automate, by the name it shows there. Defaults are what a fresh instance starts with (reset brings them back). The store's effects follow on the next pages.

Spectral smear (blue)

Name in your DAW	ID	Range	Default
Spectral Mix	`sgMix`	0 % to 100 %	35 %
Spectral Freeze	`sgFreeze`	off / on	Off
Spectral Tilt	`sgTilt`	-6.0 dB/oct darker to +6.0 dB/oct brighter	flat

Pitch (yellow)

The scale is on the card.

Name in your DAW	ID	Range	Default
Pitch Shift	`psShift`	-24.0 st to +24.0 st	+7.0 st
Pitch Feedback	`psFeedback`	0 % to 95 %	35 %
Pitch Mix	`psMix`	0 % to 100 %	30 %
Scale	`psScale`	Major (Ionian), Natural Minor (Aeolian), Harmonic Minor, Melodic Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Major Pentatonic, Minor Pentatonic, Blues, Phrygian Dominant, Whole Tone, Diminished (Whole-Half), Chromatic, The Well-Tuned Piano (La Monte Young), Raga Bhairav, Raga Yaman, Raga Bhairavi, Raga Todi, Raga Marwa, Raga Purvi, Raga Kafi, Raga Malkauns, Raga Darbari Kanada, Raga Bhupali, Raga Desh, Raga Puriya Dhanashri, 22 Shruti, Maqam Rast, Maqam Bayati, Maqam Saba, Slendro, Pelog, Harmonic Series 8-16, Partch 43-Tone, Bohlen-Pierce (tritave), Quarter-tone (24-EDO), 19-EDO, 31-EDO	Major (Ionian)

Grains (mint)

These also shape the frozen sound balloons.

Name in your DAW	ID	Range	Default
Grain Mix	`grMix`	0 % to 100 %	40 %
Grain Size	`grSize`	10 ms to 1.50 s	120 ms
Grain Density	`grDensity`	1.0x to 24.0x	7.0x
Grain Shape	`grShape`	saw, reverse saw, bell, triangle, pulse, sine, chalk	saw
Grain Random Pan	`grPan`	0 % to 100 %	45 %
Grain Reverse	`grReverse`	0 % to 100 %	0 %
Grain Position	`grPos`	0 % back to 100 % back	50 % back
Grain Position Random	`grPosRand`	0 % to 100 %	100 %
Grain Octave	`grOctave`	-2 oct to +2 oct	0 oct
Random Octave	`grRandOct`	off / on	On
Freeze	`grFreeze`	off / on	Off
Freeze Fade	`grFade`	10 ms to 10.00 s	1.00 s

Reverb (lilac) and the loop (pink)

Name in your DAW	ID	Range	Default
Reverb Decay	`rvDecay`	0.2 s to 20.0 s	4.0 s
Reverb Mix	`rvMix`	0 % to 100 %	30 %
Main Feedback	`masterFb`	0 % to 100 %	0 %

The gumball machine (orange)

Bought at the balloon store.

Name in your DAW	ID	Range	Default
Stutter Bought	`stOwned`	off / on	Off
Stutter	`stAmount`	0 % to 100 %	0 %
Stutter Bounce	`stBounce`	0 % to 100 %	50 %
Stutter Sway	`stSway`	0 % to 100 %	0 %
Stutter Cut-up	`stCutup`	0 % to 100 %	25 %

Sound balloons, pops and the canister

Name in your DAW	ID	Range	Default
Catch From	`catchFrom`	effects, input	effects
Frozen Balloons Volume	`grainsLevel`	silent to 6.0 dB	0.0 dB
Frozen Grains Through FX	`grainsThru`	straight out to through fx	straight out
Balloon Sounds	`balloonSfx`	off / on	On
Pop Sounds Through FX	`popsThru`	off / on	Off

Weather

Name in your DAW	ID	Range	Default
Cloudy	`cloudy`	off / on	Off
Rain	`rain`	off / on	Off
Snow	`snow`	off / on	Off
Heatwave	`heat`	off / on	Off
Helium	`helium`	off / on	Off
Auto Weather	`autoWeather`	off / on	Off
Snow Fade In	`snowIn`	5 s to 5 min	5 s
Snow Fade Out	`snowOut`	5 s to 5 min	5 s
Heatwave Fade In	`heatIn`	5 s to 5 min	5 s
Heatwave Fade Out	`heatOut`	5 s to 5 min	5 s
Fog	`fog`	off / on	Off
Wind	`wind`	off / on	Off
Hail	`hail`	off / on	Off
Aurora	`aurora`	off / on	Off

Twister	`twister`	off / on	Off

Sonic rainboom

Name in your DAW	ID	Range	Default
Rainboom	`rbBoom`	off / on	Off
Rainboom Level	`rbLevel`	0 % to 100 %	70 %
Rainbow Length	`rbTail`	1.0 s to 12.0 s	5.0 s
Rainboom On Throws	`rbThrows`	off / on	On
Rainboom On Hits	`rbHits`	off / on	Off
Rainboom After Storms	`rbStorms`	off / on	Off

Bypass

Name in your DAW	ID	Range	Default
Bypass Effects	`fxBypass`	off / on	Off

The balloon store's effects

Each effect has a *Bought* switch (whether it's on the shelf), its two controls, a level (up to 200 %) and a width (from mono in the middle to wide stereo). The tuned ones follow the scale.

Reverb

Name in your DAW	ID	Range	Default
Reverb Bought	`stReverbOwned`	off / on	Off
Reverb Feedback	`stReverbA`	0.00 to 1.00	0.75
Reverb Cutoff	`stReverbB`	100 Hz to 15.00 kHz	5.00 kHz
Reverb Level	`stReverbLevel`	0 % to 200 %	100 %
Reverb Width	`stReverbWidth`	mono to stereo	50 % wide

Delay

Name in your DAW	ID	Range	Default
Delay Bought	`stDelayOwned`	off / on	Off
Delay Delay	`stDelayA`	10 ms to 1.0 s	250 ms
Delay Feedback	`stDelayB`	0.00 to 1.00	0.50
Delay Level	`stDelayLevel`	0 % to 200 %	100 %
Delay Width	`stDelayWidth`	mono to stereo	50 % wide

Disto

Name in your DAW	ID	Range	Default
Disto Bought	`stDistoOwned`	off / on	Off
Disto Drive	`stDistoA`	0.00 to 1.00	0.75
Disto Slope	`stDistoB`	0.00 to 0.99	0.75
Disto Level	`stDistoLevel`	0 % to 200 %	100 %
Disto Width	`stDistoWidth`	mono to stereo	50 % wide

Waveguide

Name in your DAW	ID	Range	Default
Waveguide Bought	`stWaveguideOwned`	off / on	Off
Waveguide Frequency	`stWaveguideA`	20.6 Hz to 494 Hz	98.0 Hz
Waveguide Fall time	`stWaveguideB`	1.0 s to 60.0 s	30.0 s
Waveguide Level	`stWaveguideLevel`	0 % to 200 %	100 %
Waveguide Width	`stWaveguideWidth`	mono to stereo	50 % wide

Complex Resonator

Name in your DAW	ID	Range	Default
Complex Resonator Bought	`stComplexResOwned`	off / on	Off
Complex Resonator Frequency	`stComplexResA`	20.6 Hz to 3.95 kHz	494 Hz

Complex Resonator Decay	stComplexResB	1 ms to 5.0 s	1.0 s
Complex Resonator Level	stComplexResLevel	0 % to 200 %	100 %
Complex Resonator Width	stComplexResWidth	mono to stereo	50 % wide

Degrade

Name in your DAW	ID	Range	Default
Degrade Bought	stDegradeOwned	off / on	Off
Degrade Bit depth	stDegradeA	2.0 bits to 32.0 bits	8.0 bits
Degrade Sr scale	stDegradeB	0.01 to 1.00	0.25
Degrade Level	stDegradeLevel	0 % to 200 %	100 %
Degrade Width	stDegradeWidth	mono to stereo	50 % wide

Harmonizer

Name in your DAW	ID	Range	Default
Harmonizer Bought	stHarmonizerOwned	off / on	Off
Harmonizer Transpose	stHarmonizerA	-12 st to +12 st	-7 st
Harmonizer Feedback	stHarmonizerB	0.00 to 1.00	0.25
Harmonizer Level	stHarmonizerLevel	0 % to 200 %	100 %
Harmonizer Width	stHarmonizerWidth	mono to stereo	50 % wide

Clipper

Name in your DAW	ID	Range	Default
Clipper Bought	stClipperOwned	off / on	Off
Clipper Threshold	stClipperA	0.00 to 0.25	0.10
Clipper Cutoff	stClipperB	100 Hz to 15.00 kHz	5.00 kHz
Clipper Level	stClipperLevel	0 % to 200 %	100 %
Clipper Width	stClipperWidth	mono to stereo	50 % wide

Phaser

Name in your DAW	ID	Range	Default
Phaser Bought	stPhaserOwned	off / on	Off
Phaser Lfo freq	stPhaserA	0.05 Hz to 10 Hz	0.30 Hz
Phaser Feedback	stPhaserB	0.00 to 1.00	0.50
Phaser Level	stPhaserLevel	0 % to 200 %	100 %
Phaser Width	stPhaserWidth	mono to stereo	50 % wide

Detuned Resonator

Name in your DAW	ID	Range	Default
Detuned Resonator Bought	stAllpassWGOowned	off / on	Off
Detuned Resonator Frequency	stAllpassWGA	20.6 Hz to 494 Hz	98.0 Hz
Detuned Resonator Detune	stAllpassWGB	0.00 to 1.00	0.50

Detuned Resonator Level	stAllpassWGLevel	0 % to 200 %	100 %
Detuned Resonator Width	stAllpassWGWidth	mono to stereo	50 % wide

Ring Mod

Name in your DAW	ID	Range	Default
Ring Mod Bought	stRingModOwned	off / on	Off
Ring Mod Frequency	stRingModA	20.6 Hz to 5.27 kHz	440 Hz
Ring Mod Follow	stRingModB	0.00 to 1.00	0.30
Ring Mod Level	stRingModLevel	0 % to 200 %	100 %
Ring Mod Width	stRingModWidth	mono to stereo	50 % wide

Presets

Open them from the card at the top left (or your DAW's preset list). Each has its own picture behind the shop.

1. Sugarcube Corner
2. Canterlot Chimes
3. Moonlight Howl
4. Smokey Mountain Squall
5. Las Pegasus Neon Loop
6. Cloudsdale Stretch
7. Still Lake
8. Slow Rainbow Falls
9. Golden Hour
10. Ponyville Morning
11. Main Street Shimmer
12. Ghastly Gorge Echo
13. Froggy Bottom Chorus
14. Rock Farm Crunch
15. Rainbow Resonators

Problems and answers

macOS says it can't be opened, or the DAW skips it

The builds aren't notarised. Run the `xattr` line from Installing, then rescan. To check the Audio Unit yourself: `auval -v aufx Fkpi Frkp`.

The sound changes on its own

Something in the weather is doing it. Clouds make the balloons inside them wander (and in the Balloon Palace the tokens too), auto weather changes the sky every few minutes, a storm's lightning lends a store effect for a while, and Trixie drops by now and then. Turn off cloudy and auto weather for a steady patch.

A rainbow or a twister came by itself

A rainbow follows any shower of 20 seconds or more that ends under a clear sky. A twister comes every few minutes while wind and cloudy are both on. Turn off the wind, or keep the clouds while the rain stops, to go without.

Everything has stopped or gone still

It's snowing: snow freezes the sound until you click the snowflake again. Also check the bypass on the card.

A rainboom went off by itself

Right-click the rainboom cloud: on hits, after storms and on throws each set one off. Untick the ones you don't want.

A new instance doesn't start from the defaults

Every new instance starts from the last patch you had open. A saved session still opens as you saved it. Reset (top right) puts everything back to the first preset.

It's loud, or a rainboom is too much

The limiter keeps the output under -1 dBFS whatever happens. Turn the rainboom down in its menu (level).

Credits and license

This is an unofficial fan project, not affiliated with or endorsed by Hasbro. My Little Pony and its characters are trademarks of Hasbro. The character pictures and backgrounds are fan-made, as is the "Magic is Friendship" title font, and they belong to their creators. The flying Rainbow Dash is by Nightmare Moon.

The balloon-store effects are rebuilt from Olivier Bélanger's SoundGrain (the phaser replaces its flanger). The gumball machine, the ring mod and the bouncing plucks take their ideas from Alexander Mesker's thesis *Analysis and recreation of key features in selected Autechre tracks from 1998-2005* (Macquarie University, 2007).

Built with JUCE. The code is released under the GNU Affero General Public License v3, as JUCE's free licence requires. The artwork is not covered by it.

Source, releases and issues: github.com/femifleming/PinkiePiesBalloonShop.